

Practical: Windows Server Security, Backup and Performance Monitoring

Task 1: Configure Local Security Policy

Aim

To enforce password complexity requirements using Local Security Policy.

Steps

1. Log in to Windows Server as **Administrator**.
2. Press **Windows + R**, type:

secpol.msc

3. Press **Enter**.
4. Navigate to:

Security Settings → Account Policies → Password Policy

5. Double-click **Password must meet complexity requirements**.
6. Select **Enabled**.
7. Click **Apply** and then **OK**.
8. (Optional) Configure:
 - **Minimum password length:** 8 characters
 - **Maximum password age:** 90 days
 - **Minimum password age:** 1 day
9. Close the **Local Security Policy** window.

Result

Password complexity policy was successfully enabled.

Task 2: Enable Windows Defender and Schedule Daily Quick Scan

Aim

To enable Windows Defender and schedule a daily quick scan.

Steps

Enable Windows Defender

1. Open **Windows Security**.
2. Go to **Virus & threat protection**.
3. Ensure **Microsoft Defender Antivirus** is enabled.

Schedule Daily Scan

1. Open **Task Scheduler**.
2. Navigate to:

Task Scheduler Library → Microsoft → Windows → Windows Defender

3. Locate **Windows Defender Scheduled Scan**.
4. Right-click it and select **Properties**.
5. Open the **Triggers** tab.
6. Click **New**.
7. Select:
 - **Begin the task:** On a schedule
 - **Daily**
 - **Time:** 10:00 AM (or any desired time)
8. Click **OK**.
9. Click **Apply** and **OK**.

Result

Windows Defender is enabled and a daily quick scan is scheduled.

Task 3: Configure Windows Server Backup

Aim

To create a scheduled backup of the **ProjectFiles** folder.

Steps

1. Open **Server Manager**.

2. Select **Add Roles and Features**.
3. Install the **Windows Server Backup** feature.
4. Open **Windows Server Backup**.
5. Click **Local Backup**.
6. Select **Backup Schedule**.
7. Click **Custom**.
8. Click **Add Items**.
9. Select the folder:

C:\ProjectFiles

10. Choose **Daily Backup**.
11. Select the backup destination:

D:\Backup

(or another drive)

12. Finish the wizard.

Result

The **ProjectFiles** folder is scheduled for daily backup.

Task 4: Monitor CPU and Memory Using Performance Monitor

Aim

To monitor CPU and memory usage for 30 minutes.

Steps

1. Press **Windows + R**.
2. Type:

perfmon

3. Press **Enter**.
4. Open **Performance Monitor**.

5. Expand:

Data Collector Sets → User Defined

6. Right-click **User Defined**.

7. Select:

New → Data Collector Set

8. Name it:

Server Performance Monitor

9. Select **Create manually**.

10. Choose **Performance Counter**.

11. Add the following counters:

- **Processor → % Processor Time → _Total**
- **Memory → Available MBytes**

12. Set **Sample Interval**: 15 seconds.

13. Finish the wizard.

14. Start the Data Collector Set.

15. Let it run for **30 minutes**.

16. Stop the collector.

17. Right-click the report and choose:

- **Save As CSV**, or
- Take a screenshot.

Result

CPU and Memory usage were successfully monitored for 30 minutes.

Task 5: Report (200–300 Words)

Windows Server Security, Backup and Performance Monitoring Report

In this practical assignment, I configured several important administrative features in Windows Server to improve system security, reliability, and performance monitoring. First, I enabled the password complexity policy using the Local Security Policy console. This ensures that all user passwords must contain a combination of uppercase letters, lowercase letters, numbers, and special characters. This configuration helps protect the server from weak passwords and unauthorized access.

Next, I verified that Microsoft Defender Antivirus was enabled and configured a daily quick scan using Task Scheduler. Scheduling automatic scans helps detect malware regularly without requiring manual intervention, improving overall system security.

I also installed the Windows Server Backup feature and created a scheduled backup for the **ProjectFiles** folder. The backup was configured to save on a different drive, ensuring that important data can be restored if files are accidentally deleted or the primary drive fails.

Finally, I used Performance Monitor to create a Data Collector Set for monitoring CPU usage and available memory. The system collected performance data for 30 minutes, allowing me to analyze server resource utilization and identify any potential performance issues.

During the practical, I faced minor challenges in locating the correct Performance Monitor counters and configuring the backup schedule. I resolved these issues by carefully following the Windows Server management tools and verifying each configuration step. Overall, the practical helped me understand how Windows Server security policies, scheduled backups, and performance monitoring contribute to maintaining a secure and reliable server environment.